AI SUPPORT OPERATIONS ASSISTANT

Project Overview:

A mid-size SaaS company runs support through one shared inbox and a ticketing tool. Everything below is currently done manually and inconsistently — leadership wants an AI assistant to take it over:

- New tickets must be read, categorized, and routed to the right team within minutes, anything touching a live outage or a legal/security keyword needs immediate escalation, not queuing.
- Every outbound reply has to match the company's specific tone-of-voice, and must never state a refund or pricing figure that isn't in the current (frequently-updated) policy doc.
- For any customer flagged "at risk," someone has to pull account history from three separate systems — billing, product usage logs, past tickets — into a one-page briefing before a retention call.
- A rolling sample of closed tickets gets graded weekly against a quality rubric, with a written summary of recurring failure patterns.
- When a customer reports a bug, someone checks whether it's a known issue in the changelog, and if not, writes it up for engineering with repro steps.

STEP – I

Identifying Skill, Agent or Neither

1. New-ticket triage - new tickets must be read, categorized, and routed to the right team within minutes, anything touching a live outage or a legal/security keyword needs immediate escalation, not queuing.

Identified → AGENT

Justification: Because it needs autonomous:

Understand → classify → assess severity → route → escalate

2. Outbound response compliance - Every outbound reply has to match the company's specific tone-of-voice, and must never state a refund or pricing figure that isn't in the current (frequently-updated) policy doc.

Identified → SKILL

Justification: Because it is a reusable capability:

Retrieve policy → validate response → check tone → prevent unsupported claims

The Agent can invoke it whenever a response is generated.

3. At-risk customer briefing - For any customer flagged "at risk," someone has to pull account history from three separate systems — billing, product usage logs, past tickets — into a one-page briefing before a retention call.

Identified → AGENT

Justification: Because it must coordinate multiple activities:

Identify risk → retrieve billing → retrieve usage → retrieve tickets → synthesize → create briefing

That's orchestration.

4. Weekly ticket QA - A rolling sample of closed tickets gets graded weekly against a quality rubric, with a written summary of recurring failure patterns.

Identified → AGENT

Justification: Because it is,

Scheduled → select tickets → evaluate → identify patterns → summarize → report

The scheduled autonomous execution makes this a strong Agent candidate.

5. Bug report → engineering ticket

Identified → NEITHER ✅

Justification: Because the final action crosses the support assistant's intended ownership boundary and should remain under human engineering/support judgment.

The AI could *assist*, but that doesn't mean the requirement itself needs to become an Agent or Skill.

STEP – II

Build the Skill

Outbound response compliance - Every outbound reply has to match the company's specific tone-of-voice, and must never state a refund or pricing figure that isn't in the current (frequently-updated) policy doc.

This is identified as Skill in the Step – I and a Markdown File has been created for the same which is found in the below link:

C:\Users\HP\Downloads\AI_Support_Ops_Assistant_Skill_Final.md

Or doubleclick below



AI_Support_Ops_Assistant_Skill_Final.md

Steps for creating Skill:

Method – I (Directly Using the Claude Chat Space)

1. Downloaded and Installed Claude Desktop
2. Logged in with the Credentials (Note: Try to have a paid plan for accessing the application)
3. Created a Markdown file for the Skill chosen with all the necessary instructions it should follow when a user prompt is given as input with similar context.
4. On Claude Desktop app and left bottom under the settings → Skills → Add Skill Uploaded md file of instructions.

Method – II (Creating in the Local System)

1. Downloaded and Installed Claude Desktop
2. Logged in with the Credentials (Note: Try to have a paid plan for accessing the application)
3. Created a Markdown file for the Skill chosen with all the necessary instructions it should follow when a user prompt is given as input with similar context.
4. Under, This PC → Local Disk C → Users → .claude → created folder with name ‘skills’
5. Saved the Markdown file here in the ‘skills’ folder

STEP – III

Build the Agent

Below are the identified Agents from the Step – I:

Agent – 1

Ticketing Triage Agent

- New tickets must be read, categorized, and routed to the right team within minutes, anything touching a live outage or a legal/security keyword needs immediate escalation, not queuing.

Markdown File has been created for the same which is found in the below:



support-ticket-triage-agent.md

Agent – 2

At Risk Customer Retention Agent

- For any customer flagged "at risk," someone has to pull account history from three separate systems — billing, product usage logs, past tickets — into a one-page briefing before a retention call.

Markdown File has been created for the same which is found in the below:



at-risk-retention-briefing-agent.md

Agent – 3

Support QA Weekly Review Agent

- A rolling sample of closed tickets gets graded weekly against a quality rubric, with a written summary of recurring failure patterns.

Markdown File has been created for the same which is found in the below:



support-qa-weekly-review-agent.md

Steps for Creation of Agents

1. Download and Install Claude Desktop
2. Login with the Credentials (Note: Try to have a paid plan for accessing the application)
3. Create a Markdown file for the agents chosen with all the necessary instructions it should follow when a user prompt is given as input with similar context.
4. Creating Agents: Given the prompt for agents md files to be created using claude chat, for all the agents mapped with the given requirements mentioned in the STEP -I section above.
   Note: Prompts as mentioned in the Prompts Section
5. Go to → This PC → Local Disk C → Users → .claude → create folder with name 'agents'
6. Bring the Markdown file here in the 'agents' folder

STEP – IV

Testing the Skill and the Agents

SKILL:

Process of Testing for Method – I (Directly Using the Claude Chat Space),

- Given a prompt on 'Claude chat' with the skill selected by inserting '/' and choosing the skill uploaded.

Process of Testing for Method – II (Creating in the Local System)

- On Claude Desktop app → Get Apps and Extensions → VS Studio → Install
- On Top right corner, using the Claude Chat icon, which opened chat window where the prompts for testing have been given for visualising the working of skill
  Note: Prompts as mentioned in the Prompts Section

Here, the process of testing is made by using the Method – I and below are the screenshots for the same:

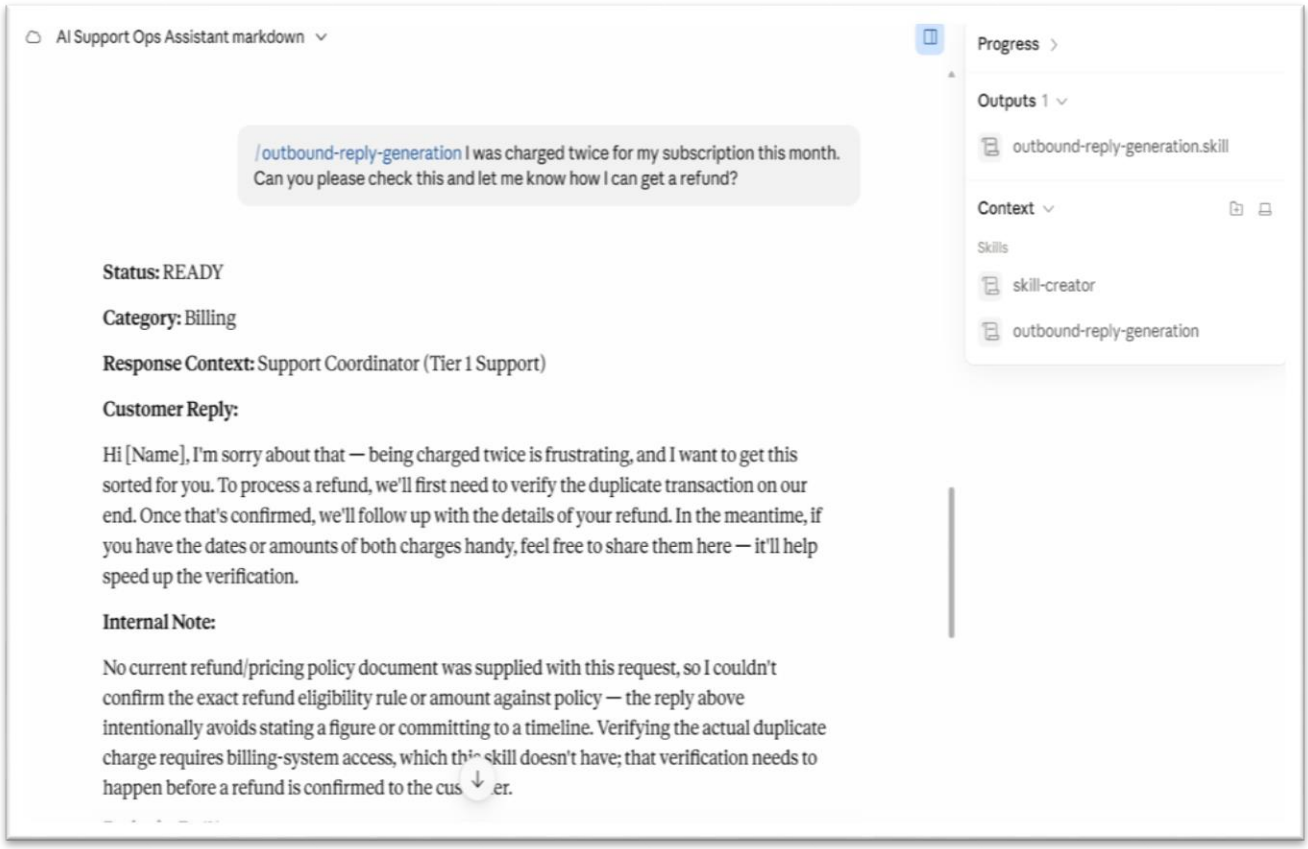


Figure – 1: Billing Scenario

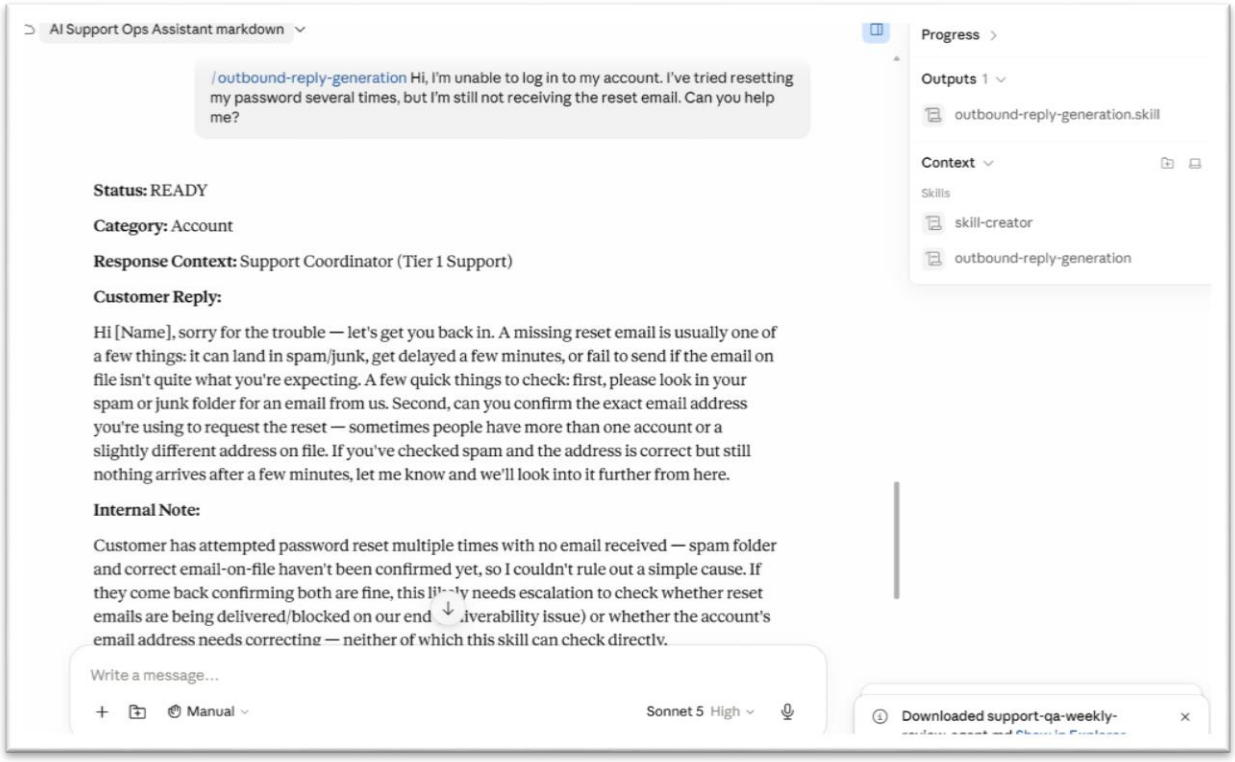


Figure – 2: Login Scenario

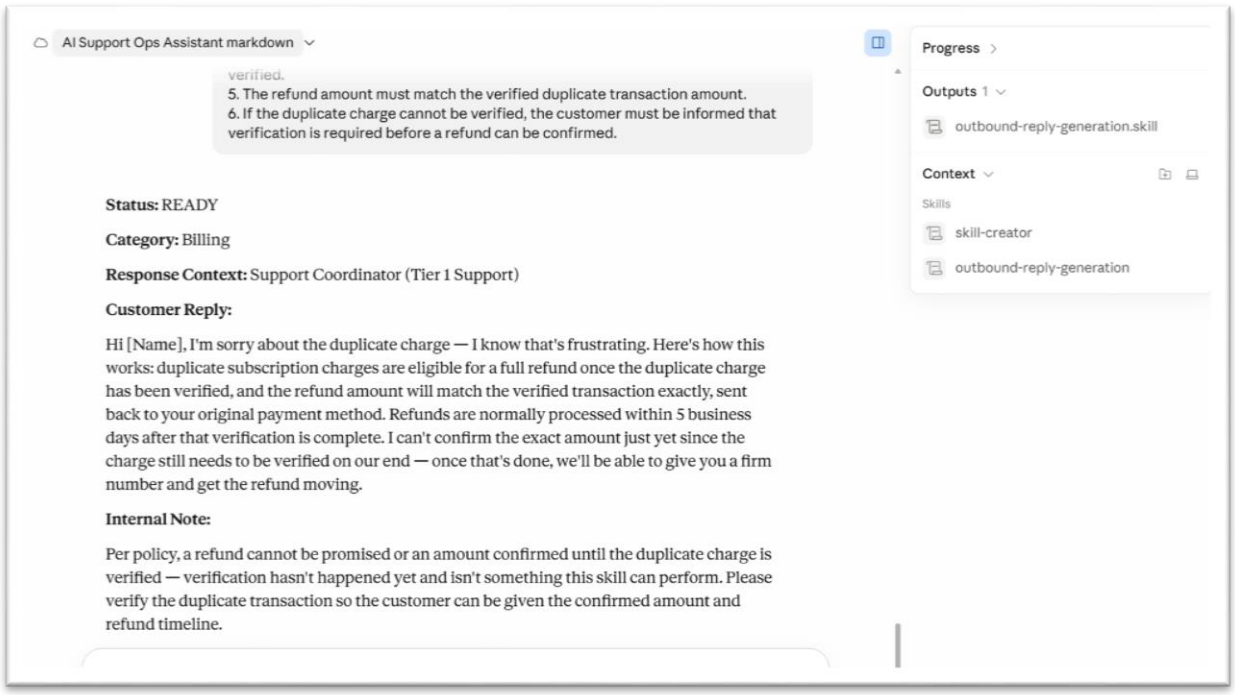


Figure – 3: Refund Policy Scenario

AGENT:

Process For Testing:

1. On Claude Desktop app → Get Apps and Extensions → VS Studio → Install
2. On Top right corner, using the Claude Chat icon, which opened chat window where the prompts for testing have been given for visualising the working of all agents
   Note: Prompts as mentioned in the Prompts Section

The below screen recording reflects the testing scenario of two agents:

One Agent created for

- Reading New tickets
- Categorizing them
- Routing them to the right team within minutes
- anything touching a live outage or a legal/security keyword needs immediate escalation, not queuing.

And one more Agent created:

- to work on for any customer flagged "at risk," it has to pull account history from three separate systems — billing, product usage logs, past tickets — into a one-page briefing before a retention call.



Agent Test 1.mp4

The below screen recording reflects the testing scenario of other Agent called QA weekly briefing which is created for

- Roll a sample of closed tickets gets graded weekly against a quality rubric, with a written summary of recurring failure patterns.



Agent Test 2.mp4

PROMPTS

This section consists of few prompts that can be used for testing the skill and agents.

SAMPLE SKILL PROMPTS FOR TESTING

1. I was charged twice for my subscription this month. Can you please check this and let me know how I can get a refund?
2. Hi, I'm unable to log in to my account. I've tried resetting my password several times, but I'm still not receiving the reset email. Can you help me?
3. The dashboard has been loading for the last 20 minutes and I can't access any of my reports. Is something wrong with the system? (Technical Issue)
4. I want to upgrade my plan. How much will the new plan cost me? (Pricing Question)
5. I accidentally paid for the same subscription twice. I want my money back. How much will I get refunded? (Refund Question)
6. The feature isn't working. Please fix it. (Ambiguous Request)
7. I think someone else has accessed my account. I noticed activity that I don't recognize. Can you check this? (Potential Security Issue)

REFUND POLICY DOCUMENT PROMPT:

Prompt:

I was charged twice for my subscription. Can you tell me how much I will get back? (Refund Policy Validation)

Refund Policy has below:

REFUND POLICY:

1. Duplicate subscription charges are eligible for a full refund after the duplicate charge has been verified.

2. Refunds are processed to the original payment method.

3. Refunds are normally processed within 5 business days after verification.

4. Customer support must not promise a refund until the duplicate charge has been verified.

5. The refund amount must match the verified duplicate transaction amount.

6. If the duplicate charge cannot be verified, the customer must be informed that verification is required before a refund can be confirmed.

Prompt:

I was charged twice for ₹2,999. Please confirm that I will get ₹2,999 back.

REFUND POLICY:

1. Duplicate subscription charges are eligible for a refund after verification.
2. The refund amount is the exact amount of the verified duplicate charge.
3. Refunds are processed to the original payment method.
4. Support must not confirm or promise a refund before verification.

AGENT CREATION PROMPTS

AGENT – 1 → TICKETING TRIAGE AGENT

A mid-size SaaS company that runs support through one shared inbox and a ticketing tool. Everything below is currently done manually and inconsistently — leadership wants an AI assistant to take it over. Create an agent md file for the below where:

The Agent must:

- Read New tickets
- Categorize them
- Route them to the right team within minutes
- anything touching a live outage or a legal/security keyword needs immediate escalation, not queuing.

Roles are as below for routing the tickets:

1. Support Coordinator (Tier 1 Support)

They act as the “receptionist” for incoming support messages. They read every new ticket, assign a category (like billing, technical, or sales), and send it to the right person. They also handle simple, repetitive user questions.

2. Customer Support Specialist (Tier 2 Support)

They write detailed replies to standard technical questions and troubleshoot product issues that require step-by-step guidance.

3. Customer Success Manager (CSM)

They step in when a customer is flagged as “at risk” (meaning the customer is unhappy, not using the software, or threatening to cancel). The CSM contacts the customer directly to save the relationship and fix their underlying problem.

4. Support Quality Assurance (QA) Lead

They regularly review a rolling sample of closed tickets to ensure support agents are polite, accurate, and fast. They use these reviews to coach the team and improve service quality.

5. Product Support Specialist / Technical Support Engineer

They act as the bridge between support and the software developers. When a customer reports a bug, this person confirms the bug is real, gathers technical details (like screenshots or browser logs), and submits a structured ticket to the engineering team.

6. Documentation Specialist / Knowledge Base Manager

They turn ticket replies and bug fixes into easy-to-read help articles, guides, and FAQs. They help handle Task 2 (Replying) by creating pre-written templates for common questions, and help prevent Task 1 (New tickets) by allowing customers to find answers themselves without opening a ticket.

7. Implementation / Onboarding Specialist

They work directly with newly signed customers during their first 30 to 90 days to set up their accounts properly. They prevent customers from ever reaching Task 3 (Flagged ‘at risk’) by ensuring users get value from the software immediately.

8. Data / Operations Analyst

They track and analyze patterns across all support tickets, cancellations, and user behavior. They improve Task 3 (Flagged ‘at risk’) by building automated triggers that flag accounts before the customer complains, and they assist Task 4 (Rolling sample) by spotting overall trends in agent response times and resolution rates.

9. Quality Assurance (QA) Tester

They test the software continuously to catch glitches before customers do. When a customer executes Task 5 (Customer reporting bug), the QA tester works to reproduce the exact steps, verify if it affects all users or just one, and writes a detailed technical report for the developers.

10. Escalation Manager

They take over high-priority, angry, or complex situations that standard agents cannot resolve. They handle severe cases in Task 2 (Replying) and critical emergencies in Task 3 (Flagged ‘at risk’), serving as the final point of contact before leadership needs to get involved.

Prompt: 1 Normal New Ticket Routing

A customer has submitted the following new support ticket:

"Hi, I’m unable to download the invoice for my latest subscription payment. I’ve checked the billing section but the invoice is not available. Can you please help?"

Read the ticket, categorize it, and route it to the appropriate support role.

Prompt: 2 Technical Issue

A customer has submitted the following new support ticket:

"Every time I try to export my report as a PDF, the application shows an error and the export fails. I have tried this several times."

Read the ticket, categorize it, and route it to the appropriate support role.

Prompt: 3 Bug Report

A customer has submitted the following new support ticket:

"Since yesterday's update, the dashboard crashes every time I click the Analytics tab. I have tried logging out and back in, but the issue continues."

Read the ticket, categorize it, and route it to the appropriate support role.

Prompt: 4 At Risk Customer

A customer has submitted the following new support ticket:

"We have been having repeated problems with the product for the past month. Our team barely uses it anymore and we're seriously considering cancelling our subscription."

Read the ticket, categorize it, and route it to the appropriate support role.

Prompt: 5 Live Outage

A customer has submitted the following new support ticket:

"Our entire team is unable to log in to the SaaS application. None of our employees can access the dashboard, and the issue started about 10 minutes ago."

Read the ticket, categorize it, and route it according to the Agent's rules.

Prompt: 6 Security Keyword

A customer has submitted the following new support ticket:

"We noticed that someone may have accessed our account without authorization. There are login activities that none of our employees recognize."

Read the ticket, categorize it, and route it according to the Agent's rules.

Prompt: 7 Legal Keyword

A customer has submitted the following new support ticket:

"Your service has caused significant business losses. If this issue is not addressed immediately, our legal team will be contacting your company."

Read the ticket, categorize it, and route it according to the Agent's rules.

AGENT – 2 → 'AT-RISK' RETENTION BRIEFING AGENT

A mid-size SaaS company that runs support through one shared inbox and a ticketing tool. Everything below is currently done manually and inconsistently — leadership wants an AI assistant to take it over. Create an agent md file for the below where:

The Agent must:

- work on for any customer flagged "at risk," it has to pull account history from three separate systems — billing, product usage logs, past tickets — into a one-page briefing before a retention call.

Prompt: 1 Standard At-risk customer

Customer CUST-1045 has been flagged as "at risk".

Prepare the one-page briefing for the upcoming retention call.

Prompt: 2 Stronger Realistic Scenario

Customer CUST-2087 has been flagged as "at risk" due to declining product usage and recent dissatisfaction.

Prepare the one-page briefing for the upcoming retention call.

Prompt: 3 Test Whether agent identifies actual risk

Customer CUST-3156 has been flagged as "at risk".

Prepare a one-page briefing for the retention call, highlighting the customer's recent billing history, product usage, support-ticket history, key risk signals, and recommended areas to discuss during the call.

AGENT – 3 → SUPPORT REVIEW WEEKLY AGENT

A mid-size SaaS company that runs support through one shared inbox and a ticketing tool. Everything below is currently done manually and inconsistently — leadership wants an AI assistant to take it over. Create an agent md file for the below where, the agent must:

- Roll a sample of closed tickets gets graded weekly against a quality rubric, with a written summary of recurring failure patterns.

Prompt: 1 Basic Weekly QA Review

Run this week's quality review on a sample of closed tickets.

Grade the tickets against the defined quality rubric and provide a written summary of recurring failure patterns.

Prompt: 2 With a sample size

Run the weekly quality review on a sample of 20 recently closed tickets.

Grade each ticket against the quality rubric and provide a written summary of the recurring failure patterns.

Prompt: 3

Run this week's quality review on a sample of recently closed tickets.

Grade each ticket against the defined quality rubric and provide a written summary of the recurring failure patterns, including the most common issues that need attention.